Week 7 Solution

NPTEL PSOSM Course (Privacy and Security in Online Social Media)

Question 1

Consider the following statement: *"Hello everyone!!!, Are you enjoying psosm course --- and learning new skills."* Choose the correct code to convert it to an un-punctuated string.

Option (a)

```
punctuations = '''!()-[]{};:'"\,<>/?@#$%^&*_~'''
no_punct = ""
for char in my_str:
    if char not in punctuations:
        no_punct = no_punct + char
```

Option (b)

```
punctuations = '''!()[]{};:'"\,<>./?@#$%^&*_~'''
no_punct = ""
for char in my_str:
    if char not in punctuations:
        no_punct = char
```

Option (c) ✓

```
punctuations = '''!()-[]{};:'"\,<>./?@#$%^&*_~'''
no_punct = ""
for char in my_str:
    if char not in punctuations:
        no_punct = no_punct + char
```

Option (d)

None of these

Solution

The string contains four kinds of punctuation that must be removed: exclamation marks (!!!), a comma (,), hyphens (---) and a full stop (.). A correct program needs two things:

- **A complete punctuation set** that contains every one of these characters, especially '-' and '.'.
- **Accumulation:** each kept character must be appended, no_punct = no_punct + char. Writing no_punct = char overwrites the result every time.

Checking each option:

Option	Has '-' ?	Has '.' ?	Logic / Result
(a)	Yes	No	Appends correctly, but the final '.' survives
(b)	No	Yes	no_punct = char keeps only the last character 's'
(c)	Yes	Yes	Appends correctly; all punctuation removed

Running option (c) (verified in Python) gives:

```
Hello everyone Are you enjoying psosm course  and learning new skills
```

(The double space between 'course' and 'and' remains because only the hyphens were removed, not the spaces around them.)

Answer: Option (c).

Question 2

Consider the following code:

```
from collections import Counter
x = [2, 1, 3, 4, 5, 2]
token = ['social', 'media', 'nptel', 'nltk', 'python', 'tutorial']
pairs = Counter(dict(zip(token, x))).most_common(2)
```

Which of the following is the correct option?

(a) This code is used to retrieve an ascending sorted list having two tuples.

(b) This code is used to retrieve a descending sorted list having two tuples. ✓

(c) This code retrieves two tuples in a sorted or unsorted manner.

(d) This code returns a logical error.

Solution

Step 1: zip(token, x) pairs each word with the number in the same position, and dict() turns the pairs into a dictionary:

Key (token)	Value (count)
social	2
media	1
nptel	3
nltk	4
python	5
tutorial	2

Step 2: Counter(...) treats the values as counts.

Step 3: most_common(2) returns a list of the 2 (key, count) tuples with the highest counts, ordered from highest to lowest.

```
pairs = [('python', 5), ('nltk', 4)]
```

The output is a list of two tuples sorted in descending order of count (5 then 4). The code runs without error, and the order is always guaranteed, so options (a), (c) and (d) are wrong.

Answer: This code is used to retrieve a descending sorted list having two tuples.

Question 3

Which of the following are not stopwords in the following list?

```
stopwords = ["i", "me", "my", ... "what", "which", ... "did", "doing", "apple", "a", "an",
"the", "and", "but", "if", "or", "because", "as", "until", "while", ... "under", "again",
"further", "then", ... "should", "now"]
```

(Full list as given in the question.)

(a) All of the above are stopwords.

(b) Further, again, apple

(c) While, apple

(d) apple ✓

Solution

Stopwords are very common function words (pronouns, articles, prepositions, conjunctions, auxiliary verbs) that carry little meaning on their own. They are removed during text preprocessing so that analysis focuses on content words. The list in the question is the standard English stopword list (as in NLTK), with one word inserted that does not belong.

Checking each word in the options:

Word	Type of word	Stopword?
further	Adverb / connector	Yes, a standard stopword
again	Adverb	Yes, a standard stopword
while	Conjunction	Yes, a standard stopword
apple	Noun (a content word)	No: a meaningful noun, inserted into the list

'apple' names a real thing (a fruit or a company) and carries meaning, so removing it would lose information. It is not part of the standard NLTK stopword list; notice it sits between "doing" and "a", where the original list has no such word. 'further', 'again' and 'while' are genuine stopwords, so options (b) and (c) wrongly include them, and (a) is wrong because 'apple' is not a stopword.

You can confirm this in Python:

```
from nltk.corpus import stopwords
sw = set(stopwords.words('english'))
print('apple' in sw)    # False
print('further' in sw)  # True
```

Answer: apple.

Question 4

Which elements in the below sample mail make it potentially a phishing email, as discussed in the lecture?



(a) Urgency in subject

(b) Regrettable information in the body

(c) Potential phishing link along with body

(d) All of the above ✓

Solution

Phishing emails pretend to be from a trusted brand (here, eBay) to trick the reader into revealing credentials. This mail shows all three classic warning signs:

- **Urgency in subject:** the subject line reads 'eBay urgent security notice', pushing the reader to act before thinking.
- **Regrettable information (fear):** the body says the account could be suspended or terminated within 24 hours and threatens loss of access, creating anxiety.
- **Phishing link:** the reader is asked to click a link and re-enter account information. The sender address is also an unrelated university mailing list sending 'on behalf of' eBay, which is a further red flag.

Since all three elements are present and each is a lecture-listed phishing cue, the answer is all of them.

Answer: All of the above.

Question 5

Which of the following is the correct statement as discussed in the lecture?

(a) Younger participants were more vulnerable to phishing attacks. ✓

(b) Older participants were more vulnerable to phishing attacks.

(c) Younger and older participants were equally vulnerable to phishing attacks.

(d) No one is vulnerable to phishing attacks.

Solution

The lecture discusses user studies on who falls for phishing (for example, the study by Sheng et al., 'Who Falls for Phish?', CHI 2010). Participants aged 18 to 25 fell for phishing emails more often than older age groups. The reasons suggested were:

- Less exposure to training and awareness material about phishing.
- Less experience with online fraud and fewer years of internet use in risky contexts.
- A tendency to act quickly and trust familiar-looking brands.

Option (d) is clearly false, and the study did not find equal vulnerability, so (b) and (c) are also wrong.

Answer: Younger participants were more vulnerable to phishing attacks.

Question 6

In a scientific experiment, which group refers to manipulating the independent variable?

(a) Observational group

(b) Comparison group

(c) Control group

(d) Experimental group ✓

Solution

In a controlled experiment, the researcher changes one factor (the independent variable) and measures its effect (the dependent variable).

- **Experimental (treatment) group:** receives the manipulation. Example: users who receive anti-phishing training.
- **Control group:** does not receive the manipulation and serves as a baseline. Example: users who get no training.

Comparing the two groups shows whether the manipulation caused a change. An observational study does not manipulate anything at all. So the group in which the independent variable is manipulated is the experimental group.

Answer: Experimental group.

Question 7

Which of the following is/are the advantages of having privacy policies?

(a) To help users make informed decisions with the information that is presented to them

(b) To help individuals avoid regrettable online disclosures

(c) Both a and b are correct ✓

(d) None of the above

Solution

A privacy policy explains what data a platform collects, how it is used, who it is shared with and how long it is kept.

- **(a) Informed decisions:** knowing these facts lets a user decide whether to sign up, which permissions to grant and what settings to choose.
- **(b) Avoiding regrettable disclosures:** when users understand who can see their posts and data, they are less likely to share something they later regret (for example, posting publicly what they meant only for friends).

Both statements are benefits, so the answer is option (c).

Answer: Both a and b are correct.

Question 8

Which of the following nudges is depicted in the figure below?



(a) Timer nudge

(b) Picture nudge ✓

(c) Sentiment nudge

(d) All of the above

Solution

Privacy nudges are small interface changes that gently remind users of the consequences of posting, without blocking them (Wang et al., Facebook nudge study). The three nudges discussed in the lecture are:

Nudge	What it does
Picture (audience) nudge	Shows profile pictures of people who will be able to see the post, plus a note such as 'These people and ANYONE ON THE INTERNET can see your post'
Timer nudge	Adds a short countdown delay before a post is published, with a chance to cancel or edit
Sentiment nudge	Warns the user when the text of the post appears negative or offensive

The figure shows a row of profile photos under the post box together with the audience warning. There is no countdown and no sentiment warning, so only the picture nudge is depicted.

Answer: Picture nudge.

Question 9

As discussed in the lecture, which of the following is/are not a social media nudge?

(a) Timer Nudge

(b) Picture Nudge

(c) Sentiment Nudge

(d) Voice Nudge ✓

Solution

As listed in the table in Question 8, the lecture covers exactly three privacy nudges: picture (audience), timer and sentiment. 'Voice nudge' is not one of them; it is a distractor.

Answer: Voice Nudge.

Question 10

Which of the following has the property of having high indegree and high outdegree on online social media?

(a) Top Link farmers ✓

(b) Random social network users

(c) Top Spammers

(d) All of the above

Solution

Recall from Week 6: indegree = number of followers, outdegree = number of accounts followed. The link farming study on Twitter (Ghosh et al., WWW 2012) compared three groups:

User type	Indegree	Outdegree	Reason
Top link farmers (social capitalists)	High	High	Follow many users and follow back almost everyone who follows them
Spammers	Low	High	Follow many accounts, but few genuine users follow them back
Random users	Low	Low	Ordinary users with modest follower and following counts

Only the top link farmers have both a high indegree and a high outdegree, because their whole strategy is to follow many accounts and reciprocate follows (reciprocity). Spammers fail to attract many followers, so their indegree stays low.

Answer: Top Link farmers.

Answer Key

Q.No.	Correct answer
1	Option (c) (complete punctuation set with no_punct = no_punct + char)
2	Descending sorted list having two tuples

3	apple
4	All of the above
5	Younger participants were more vulnerable
6	Experimental group
7	Both a and b are correct
8	Picture nudge
9	Voice Nudge
10	Top Link farmers